



Sales Forecasting & Demand Intelligence

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Date: July 2026

Executive Business Report — Xylofy AI Internship, Week 3 & 4

Data: 9,800 transactions (2015–2018)

EXECUTIVE SUMMARY

We analyzed four years of sales data (9,800 transactions across 3 product categories and 4 regions) to build a system that **predicts future sales, detects unusual patterns, and segments products by demand behavior**. The system tested three different forecasting approaches and identified the most accurate one. Our best model's predictions are within ~10-15% of actual sales — accurate enough to inform real stocking and budgeting decisions. Key finding: **November-December drives 30%+ of annual revenue**, and overtime inventory positioning by October is critical.

TOTAL REVENUE \$2.3M	TRANSACTIONS 9,800	GROWTH TREND ↑ 15-20% YoY	MODELS TESTED 3
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KEY FINDINGS FROM DATA

- 1. Technology is the Revenue Leader**
Technology category generates the highest total revenue, with the highest average order value (\$450+). It also shows the strongest year-over-year growth, making it the top investment priority.
- 2. Strong Seasonality Drives Revenue**
November and December consistently spike across ALL 4 years — this isn't random, it's a reliable pattern. These two months alone drive 30%+ of annual revenue. January-February consistently dip.
- 3. Sales Growth is Real and Consistent**
Year-over-year sales have grown 15-20% annually. The trend component confirms this is genuine growth, not just seasonal noise. All 4 regions show positive trajectories.

3-MONTH SALES FORECAST

Month	Forecasted Sales	Confidence Range
Month 1	~\$40,000–\$65,000	Moderate confidence
Month 2	~\$45,000–\$70,000	Moderate confidence
Month 3	~\$50,000–\$80,000	Wider range (further out)

* Exact values depend on the month. Holiday months will be at the upper range.

TOP 3 ANOMALIES DETECTED

#	When	What Happened	Likely Cause
1	Nov-Dec	Sales 2-3× normal	Holiday season (Black Friday, Christmas)
2	Sep	Unexpected mid-year spike	Back-to-school/office bulk orders
3	Jan-Feb	Sharp post-holiday drop	Normal seasonal slowdown

Two independent detection methods (Isolation Forest + Z-Score) were used for reliability.

PRODUCT DEMAND SEGMENTS

Segment	Products	Stocking Strategy
High Vol, Stable	Phones, Chairs, Storage	High safety stock, bulk contracts
High Vol, Volatile	Binders, Paper, Accessories	Dynamic reorder, weekly monitoring
Growing Demand	Copiers, Machines	Increase stock quarterly
Low / Declining	Fasteners, Labels	Minimize inventory, consider dropship

3 BUSINESS RECOMMENDATIONS

- 1. Pre-Position Inventory by October**
November-December consistently drive 30%+ of annual revenue. All regional warehouses should be fully stocked by early October. Late positioning risks stockouts during the highest-revenue period of the year.
- 2. Invest in Technology Category Expansion**
Technology shows the strongest growth, highest order values, and most stable demand. Expand supplier partnerships, increase shelf space, and consider dedicated marketing. Every \$1 invested in Technology inventory yields higher returns than other categories.
- 3. Deploy Weekly Anomaly Alerts**
Implement automated alerts when weekly sales deviate >2 standard deviations from the 8-week rolling average. This catches both surges (trigger emergency restocking) and drops (investigate root causes) within 24 hours rather than waiting for monthly reports.

SYSTEM LIMITATION

This system predicts based on historical patterns and assumes the future will resemble the past. It cannot account for unprecedented events (pandemics, supply chain crises), new product launches, competitor actions, or changes in pricing strategy. The model should be retrained quarterly with fresh data and used as a decision-support tool alongside human judgment — not as a replacement for it.

 MODEL COMPARISON & METHODOLOGY

We tested three fundamentally different forecasting approaches to find the most accurate predictor:

Model	Type	How It Works (Non-Technical)	Accuracy
SARIMA	Statistical	Uses mathematical formulas based on past sales patterns and seasonal cycles	Good
Prophet	Industry Tool	Facebook's tool that automatically finds trends and holiday effects	Good
XGBoost	Machine Learning	Learns from hundreds of "features" (recent sales, month, quarter) to predict	Best

The winning model was selected purely on numbers (lowest prediction error), not personal preference. All three models agreed on the general direction of sales.

 DETAILED SEGMENT FORECASTS

We applied the best model independently to each product category and region:

Category	Current Trend	3-Month Outlook	Region	Current Trend	3-Month Outlook
Technology	↑ Strong growth	 Increasing	West	↑ Strong growth	 Increasing
Furniture	→ Stable	→ Stable	East	→ Stable growth	→ Steady
Office Supplies	→ Moderate growth	→ Moderate growth	Central	→ Moderate	→ Moderate
			South	→ Stable	→ Stable

 INTERACTIVE DASHBOARD

A **live, interactive dashboard** has been deployed where stakeholders can:

Dashboard Page	What You Can Do
Sales Overview	View yearly/monthly trends, filter by region and category
Forecast Explorer	Select any category or region and see 1-3 month predictions with accuracy metrics
Anomaly Report	View flagged unusual sales weeks with dates and values
Product Segments	See which products belong to which demand group, with stocking recommendations

Dashboard URL will be provided via the submission form (Streamlit Community Cloud deployment).

 PROJECT DELIVERABLES

File	Description
analysis.ipynb	Complete notebook with all 8 tasks, commented code, charts
train.csv	Superstore Sales dataset (9,800 rows)
vgsales.csv	Video Game Sales secondary dataset
app.py	Streamlit dashboard code (4 pages)
requirements.txt	All Python dependencies for redeployment
summary.pdf	This 2-page executive report
charts/	10 high-resolution chart images (.png)